Byeongchan Oh

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Education

Mar. 2025 ~ Present	**Seoul National University of Science and Technology** Department of Semiconductor Engineering *Advisor: Dr. Tae-Hyeon Kim* [Link] *M.S. Student*	Seoul, Korea
Mar. 2021 ~ Feb. 2025	**Seoul National University of Science and Technology** Department of Electronics and IT Media Engineering *B.S. in Electronics and IT Media Engineering*	Seoul, Korea

Research Interests

1. Ferroelectric and resistive nonvolatile memory devices for energy-efficient computing
2. BEOL-compatible thin-film devices for monolithic 3D integration
3. Compute-in-Memory/Processing-in-Memory hardware for energy-efficient AI

Skills

Semiconductor Device Fabrication, Materials Characterization, Electrical Characterization, Photomask Layout Design, Python, PyTorch, C, SmartSpice

Research Experience

Seoul National University of Science and Technology, Seoul Korea — Mar. 2024 – Present
Research Assistant, Seoultech Semiconductor Device LAB [Link]

Semiconductor Device Fabrication

- Developed and optimized baseline process recipes for newly installed fabrication equipment, including sputtering, ALD, dry/wet etching, ICP-CVD, and photolithography, supporting early-stage fab stabilization and device fabrication.
- Designed photomask layouts for LTPS TFTs, oxide TFTs, capacitors, FeTFTs, and RRAM single-cell devices, as well as simple circuits and nonvolatile memory arrays for CIM applications.
- Fabricated Hf-based ferroelectric MFM capacitors and BEOL-compatible nonvolatile memory devices, while continuing unit-process calibration and integration optimization.
- Optimized fabrication conditions for IGZO TFTs and established device parameters, including threshold-voltage tuning, for circuit-level integration. Demonstrated simple BEOL-compatible circuits, including charge-based SNN neuron circuits using IGZO TFTs.
- Currently developing a SnO-based p-type oxide semiconductor process toward sputtering-based BEOL CMOS implementation.
- Currently developing a BEOL-compatible LTPS TFT process involving a-Si deposition by ICP-CVD, laser annealing for poly-Si crystallization, and dopant activation after ion implantation, in collaboration with industry partners.

Device Characterization and Circuit Modeling

- Performed DC and pulse measurements of RRAM, IGZO TFTs, MFM capacitors, and FeTFTs, primarily using Keysight B1500A and additionally Keithley 4200A, for device verification and electrical characterization.
- Established an in-house array-level measurement setup using Keysight B1500A, 5250A switch matrix, and probe cards, and gained additional array-level measurement experience through collaborative research.
- Developed SmartSpice-based models using measured data from IGZO TFTs and NNFC 180 nm CMOS devices, and applied them to the design and simulation of neuron circuits, inverters, and FEOL–BEOL co-integrated circuits.
- Currently working on integrated system implementation combining FEOL CMOS and BEOL thin-film device platforms.

CIM/PIM Hardware and System-Level Evaluation

- Conducted neuromorphic hardware demonstrations by physically mapping fully connected layers onto RRAM

arrays and validating DNN inference performance on hardware.
- Investigated training and quantization strategies to optimize quantized neural networks for hardware deployment under device-level constraints.
- Preparing to extend this hardware demonstration framework to other device platforms, including FeTFTs, as stable device fabrication becomes available.
- Performed system-level analysis of a CTF flash-based PIM architecture operating in both current mode and capacitance mode using Python-based NeuroSim V1.4, with a focus on power efficiency and overall hardware performance.

Research Projects

1. Development of Memory Device and System Bonding Process Technology for Near-Sensor Computing Design and Implementation
2. Implementation of a Multifunctional AI Accelerator Based on Si/IGZO Monolithic 3D Integration
3. Development of a Flash-Based Feedback Transistor PIM Array and Integrated Computing Circuits for Logic Operation Optimization
4. Development of Ferroelectric-Based BEOL-Compatible Nonvolatile Memcapacitor Devices and Their Application to Ultra-Low-Power In-Memory Computing
5. Development of BEOL Process-Based Polarization-Controlled Nonvolatile Memcapacitor Arrays
6. Development of Low-Temperature Thin-Film Transistor-Based Neuron Circuits and Performance Simulation of Monolithic 3D IC Systems

Journal Publications

1. **Byeongchan Oh**, Wonbo Shim, and Tae-Hyeon Kim. "Wake-Up Effects on Improving Gradual Switching and Variation in Al_2O_3-Based RRAM." *Electronics* 2025, 14 (10), 1921. [Link]
2. **Byeongchan Oh**, Sangwook Youn, Hyungjin Kim, and Tae-Hyeon Kim. "Enhancing Post-Training Quantization Robustness in RRAM-Based CIM via σ-Clipping of Weights." *ACS Applied Electronic Materials* 2026, 8 (9), 4041–4051. [Link]
3. Dayeon Yu, Hwiho Hwang, **Byeongchan Oh**, Junmo Lee, Shimeng Yu, Tae-Hyeon Kim, and Hyungjin Kim. "A Unified Flash Memory Platform for Mode-Adaptive and Robust AI Computation." *Advanced Science*, 2026, e76619. [Link]
4. Je-Hyun Yoon, **Byeongchan Oh**, and Tae-Hyeon Kim. "Impact of an Ultrathin Al_2O_3 Interlayer on Memory Characteristics in HZO-FeFETs" *Microelectronics Journal*, 2026, 176, 107401. [Link]
5. Jaewon Sim, **Byeongchan Oh**, and Tae-Hyeon Kim. "Impact of RRAM On/Off Ratio and Variability on Accuracy-Efficiency Trade-offs in Compute-in-Memory Accelerators." Under Review.

Conference Proceedings

1. 32nd Korean Conference on Semiconductors (KCS)	Poster, 2025 (Domestic)
2. Summer Annual Conference of IEIE	Oral, 2025 (Domestic; **Best Paper Award**)
3. 25th International Conference on Electronics, Information, and Communication (ICEIC)	Oral, 2026 (International)
4. 33rd Korean Conference on Semiconductors	Poster, 2026 (Domestic)

Patent

Tae-Hyeon Kim, Byeongchan Oh, "MEMORY OPERATION APPARATUS USING WEIGHT CLIPPING AND OPERATING METHOD THEREOF", Korea, 1-1-2026-0786996-52, Jun. 2026 (patent application)

Other Experience

Teaching Assistant, S-FAB Semiconductor Process Training Program — Jan. 2026 / Jun. 2026
Provided hands-on training and instruction in photolithography.

Teaching Assistant, Creative Engineering Design Course — Mar. 2026 – Jul. 2026
Provided coding guidance to undergraduate students for course projects.